Module Title - COMMUNICATIONS AND NETWORKING SECURITY

Unit Code - B9CY103

Lecture – Arturo Vázquez Zepeda

Prepared by

ANSHIO RENIN MICHEAL ANTONY XAVIER SOOSAMMAL - 20036753

ASSIGNMENT – CA 40%

Client: Technovia Solutions
System: DNS & DHCP Server (VM-4)
Date: 22-04-2025

Contents

1. EXECUTIVE SUMMARY

This report shows the vulnerability findings which has conducted on a Linux-based server (VM4) (DNS & DHCP server) for Technovia Solutions. This assessment evaluated the system security posture using nmap scan and some authorized system access. This investigation revealed the advanced, critical vulnerabilities which includes the improperly configured DNS and DHCP services even after the server designated role. The server exposes an excessive number of unnecessary services and ports which is a major way to expand its attack surface. Some critical configuration issues were identified such as insecure DNS zone transfer settings, improperly implemented configuration files and the presence of unwarranted services like IRC, Samba, and Kerberos on a dedicated DNS & DHCP server.

These issues collectively give a severe security risk that could compromise the confidentiality, integrity and availability of network services. This report outlines the discovered vulnerabilities with supporting evidence provides targeted mitigation strategies and recommends future security improvements to strengthen Technovia Solution infrastructure security posture.

2. METHODOLOGY

Scanning Techniques:

- Initial port discovery using SYN scanning (nmap -sS)
- Service version detection (nmap -sV)
- Operating system detection (nmap -O)
- Comprehensive scanning with script automation (nmap -A)
- Vulnerability script scanning (nmap --script vuln)
- DNS-specific vulnerability scanning (nmap --script dns)
- DHCP discovery (nmap --script broadcast-dhcp-discover)
- SMB enumeration and vulnerability scanning (nmap --script smb)

System Access Verification:

- Direct login to verify configuration files
- Service status verification
- System configuration review
- Running process analysis

3. FINDINGS

ID	Service/ Port	Description	Severity	Evidence	Risk/Impact
VULN-001	DNS (Port 53)	Missing DNS service on designated DNS server. Service not found despite presence of configuration files.	Critical	systemctl status bind9 output: "Unit bind9 service could not be found." Multiple .dpkg-new configuration files exist in /etc/bind/ but are not properly implemented.	Complete failure of DNS functionality to preventing proper name resolution across the network.
VULN-002	DHCP (Ports 67/68)	Missing DHCP service on a designated DHCP server. Service not running despite package being installed.	Critical	systemctl status isc-dhcp-serve output: "Unit isc-dhcp-serve service could not be found." DHCP configuration exists but is not properly implemented.	Network clients unable to obtain IP addresses automatically, leading to network connectivity issues.
VULN-003	DNS (Port 53)	Insecure DNS zone transfer configuration allowing unrestricted transfers.	High	/etc/bind/named.conf.local contains zone "example.com" IN { type master; file "/etc/bind/db.example.com "; allow-transfer { any; }; };	Would allow unauthorized parties to download complete DNS zone information, exposing internal network structure.
VULN-004	Multiple Services	Excessive open ports including SSH (22), SMB (139/445), IRC (6666), KRB524 (4444), and numerous others.	Critical	Nmap scan shows numerous unnecessary open ports: nmap -sS -p- 192.168.233.128	Significantly expanded attack surface providing multiple entry points for attackers.
VULN-005	SSH (Port 22)	SSH revealing specific version information.	Medium	Nmap service scan shows OpenSSH 9.6p1 Ubuntu	Allows attackers to target known vulnerabilities in the specific SSH version.

VULN-006	SMB (Ports 139/445)	Samba service running with message signing not required.	High	Nmap output: smb2-security-mode: 3:1:1: Message signing enabled but not required	Potential for man-in-the-middle attacks against SMB traffic.
VULN-007	IRC (Port 6666)	IRC service running on a server designated for DNS/DHCP.	High	Nmap port scan showing IRC service on port 6666	Unnecessary service increasing attack surface and potentially allowing unauthorized communication channels.
VULN-008	Configuration Management	Multiple .dpkg-new files indicating improper configuration management.	High	ls -la /etc/bind/ shows numerous .dpkg-new files	Configuration changes not properly applied leading to security misconfiguration
VULN-009	System Services	PostgreSQL and Redis database services running on DNS/DHCP server.	Medium	systemctl list-units --type=service --state=running shows database services active	Unnecessary services might expand attack surface.
VULN-010	Kerberos (Port 4444)	KRB524 service running without proper configuration.	High	Port 4444 open with KRB524 service	Potential authentication service vulnerabilities and unnecessary attack surface expansion.
VULN-011	Industrial Protocol (Port 2222)	EtherNet/IP-1 industrial protocol exposed on server.	Critical	Nmap scan showing EtherNet/IP-1 on port 2222	Industrial protocol exposure could allow control system access
VULN-012	NetXMS (Ports 4701-4705)	NetXMS management and synchronization services running.	High	Nmap scan showing netxms-mgmt (4701) and netxms-sync (4702)	Unauthorized access to network monitoring.

VULN-013	FreeCiv Gaming (Port 5555)	FreeCiv gaming service running on server.	Medium	Port 5555 showing freeciv service	Unnecessary gaming service increasing attack surface.
VULN-014	Authentication	Xinetd service running, potentially allowing insecure authentication methods.	High	systemctl list-units shows xinetd.service running	Potential for brute force attacks and insecure access methods.
VULN-015	System Configuration	System has been minimized without security hardening.	Medium	This system has been minimized by removing packages and content that are not required on a system that users do not log into.	Important security tools and monitoring capabilities may be missing.
VULN-016	DNS Configuration	DNS blacklist configuration exists but is not implemented.	Medium	Nmap DNS script results showing incomplete DNS blacklist configuration	Insufficient spam and threat protection for DNS services.
VULN-017	File Permissions	Potentially insecure file permissions on configuration files.	High	File permission listing of configuration directories	Unauthorized modification of configuration files possible.
VULN-018	System Role Architecture	Fundamental role-based architecture violation with mixed service types.	Critical	Combined evidence of database, DNS, DHCP, IRC, and industrial services on same server	Complete violation of security best practices for server role isolation.
VULN-019	DHCP Configuration	Improperly configured DHCP scopes and options.	High	DHCP configuration showing basic settings without security controls	Potential for unauthorized DHCP servers and IP conflicts.
VULN-020	Network Services	Improper network service segmentation with critical and non-critical services mixed.	High	Network service listing showing inappropriate service combinations	Cross-contamination risk between different security domains.

4. MITIGATION PLAN

VULN-001 & VULN-002 - Missing DNS & DHCP Services:

- Properly activate the installed services: systemctl enable --now (bind9 and isc-dhcp-server)
- Moving the .dpkg-new files to their correct locations: mv /etc/bind/named.conf.options.dpkg-new /etc/bind/named.conf.options

VULN-003 - Insecure DNS Zone Transfer:

- Modify zone transfer configuration to restrict the specific authorized servers: allow-transfer with IP range
- Add proper access controls for zone transfers in the BIND configuration

VULN-004 - Excessive Open Ports:

- Implement strict firewall rules: ufw default deny incoming
- Disable and remove unnecessary services: systemctl disable [service] and apt remove [packages]

VULN-005 - SSH Version Banner Exposure:

- Configure SSH to hide version information by adding DebianBanner
- Implement SSH key-based authentication and disable password login

VULN-006 - SMB Message Signing Not Required:

- If SMB is required, we need to configure to mandate message signing by adding server signing = mandatory to /etc/samba/smb.conf
- Implement proper firewall rules to restrict SMB access to authorized networks only

VULN-007 - IRC Service Running:

- Disable or remove the IRC server

VULN-008 - Configuration Management Issues:

- Properly implement configuration files by removing .dpkg-new extensions
- Implement configuration management tools for consistent server configurations

VULN-009 - Unnecessary Database Services:

- Disable unneeded database services
- Remove database packages not required for core functionality

VULN-010 - Kerberos Service Misconfiguration:

- Remove Kerberos service if not required

- If required, restrict access to Kerberos services via firewall rules and give authentication and authorisation

VULN-011 - Industrial Protocol Exposure:

- Remove or disable the EtherNet/IP service on port 2222
- Block port 2222 in firewall rules
- Make sure industrial protocols only deployed on isolated networks

VULN-012 - NetXMS Services Exposure:

- Remove or disable NetXMS services if not explicitly required
- Implement network monitoring solution with security controls

VULN-013 - FreeCiy Gaming Service:

- Remove gaming service from production server
- Block gaming port access via firewall rules

VULN-014 - Xinetd Service Risk:

- Remove xinetd if not required
- Replace xinetd with secure alternatives or direct service configuration

VULN-015 - System Minimization Issues:

- Install security monitoring utilities: apt install auditd fail2ban
- Implement security hardening according to CIS benchmarks

VULN-016 - DNS Blacklist Configuration:

- Properly implement DNS blacklist by configuring it active configuration files
- Implement additional DNS security measures like DNSSEC

VULN-017 - File Permission Issues:

- Set correct permissions on configuration files
- Ensure proper ownership: chown root:bind /etc/bind/*

VULN-018 - Role-Based Architecture Violation:

- Redesign server architecture to follow role-based security principles
- Separate DNS/DHCP services onto dedicated servers

VULN-019 -- DHCP Configuration Issues:

- Implement proper DHCP configuration with secure options
- Configure DHCP snooping on network equipment
- Set up proper DHCP failover and high availability

VULN-020 -- Network Service Segmentation:

- Implement network segmentation strategies
- Establish proper network zones with appropriate security controls

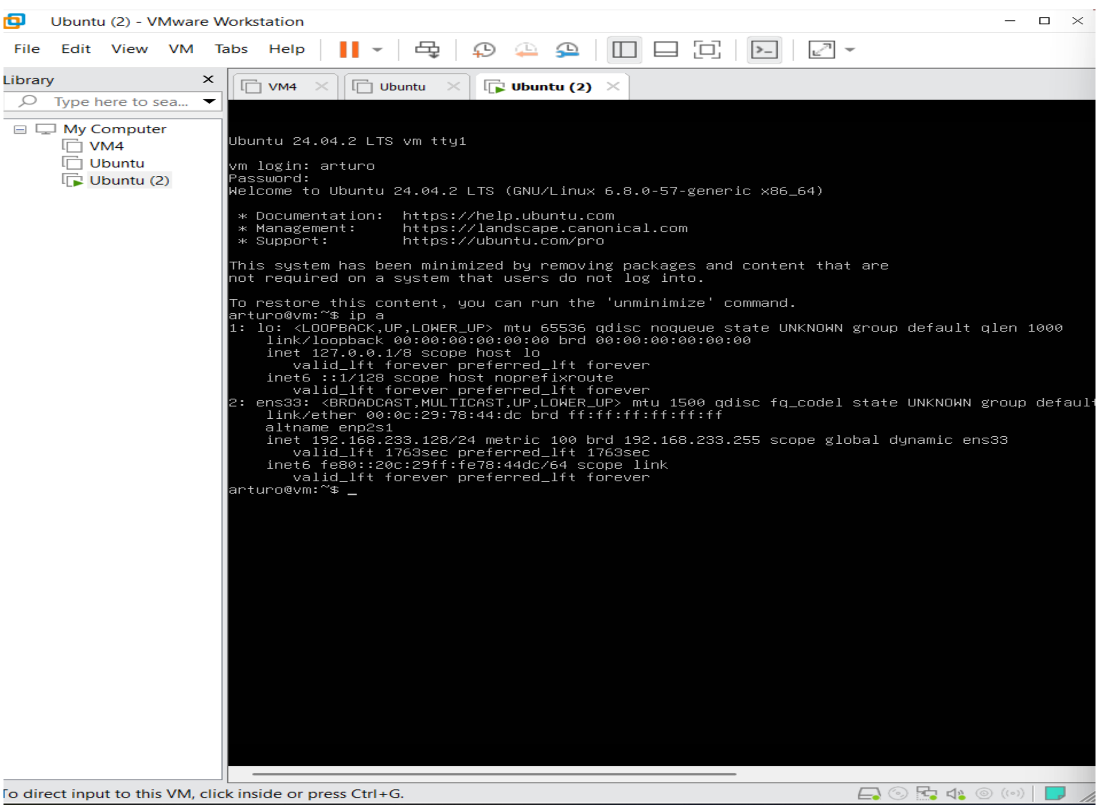
5. FUTURE RECOMMENDATIONS

- Implement role-based server architecture limiting each server to its designated function
- Use tools for infrastructure-as-code to ensure secure configurations
- Establish regular vulnerability scanning program to detect security issues
- Develop security baselines specific to each server role like DNS, DHCP
- Implement automated security patching process
- Maintain documentation of configurations and security settings
- Provide technical staff with security awareness training focusing on server hardening

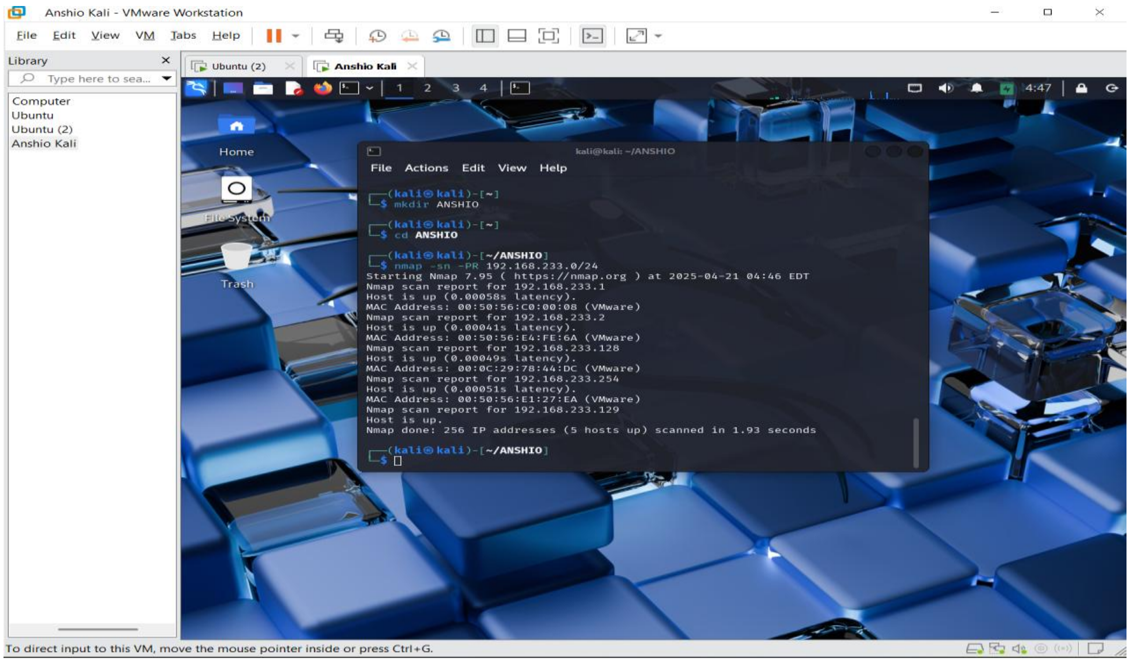
6. APPENDICES

Appendix A: Set Up

VM 4 - Ip Found: 192.168.233.128



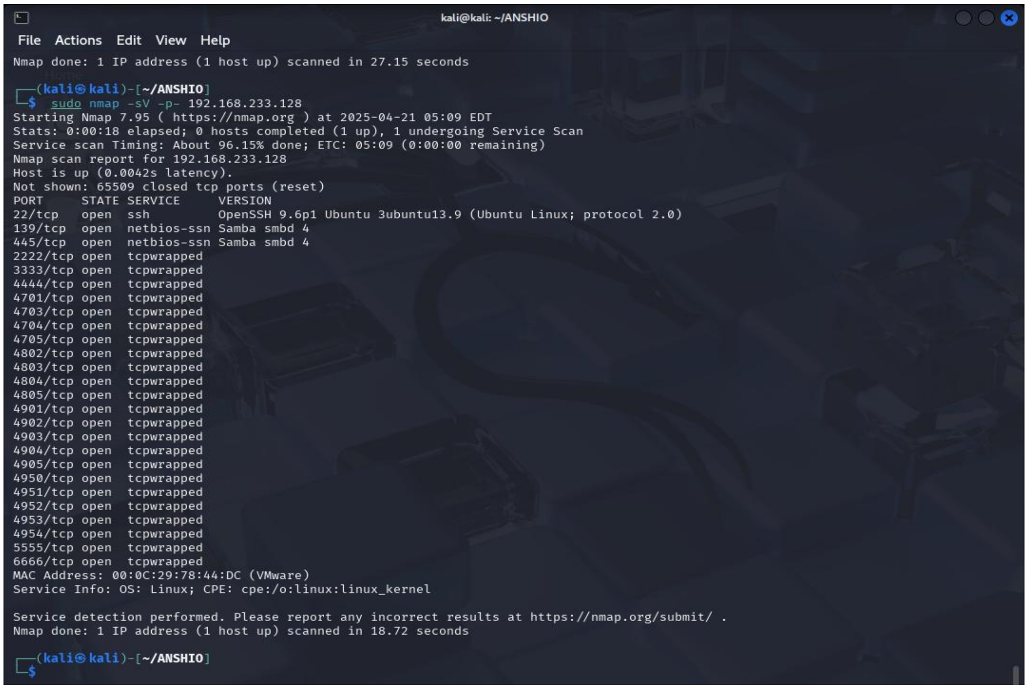
Pinging from kali



Appendix B: Nmap Scan Results

Initial Port Scan

Service Version Detection

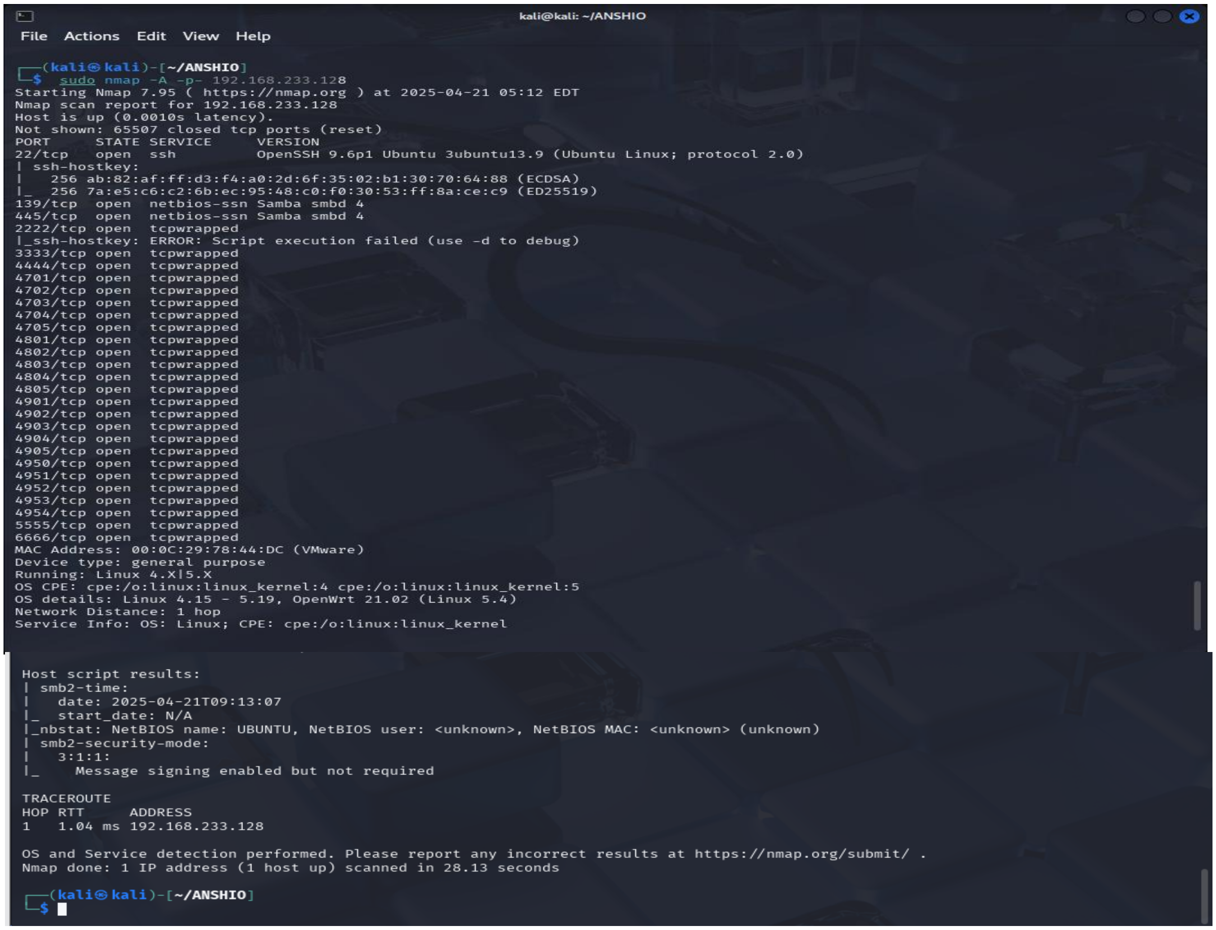


```
kali@kali: ~/ANSHIO
File  Actions  Edit  View  Help
Nmap done: 1 IP address (1 host up) scanned in 27.15 seconds

┌──(kali㉿kali)-[~/ANSHIO]
└─$ sudo nmap -sV -p- 192.168.233.128
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-21 05:09 EDT
Stats: 0:00:18 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 96.15% done; ETC: 05:09 (0:00:00 remaining)
Nmap scan report for 192.168.233.128
Host is up (0.0042s latency).
Not shown: 65509 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
22/tcp   open  ssh         OpenSSH 9.6p1 Ubuntu 3ubuntu13.9 (Ubuntu Linux; protocol 2.0)
139/tcp  open  netbios-ssn Samba smbd 4
445/tcp  open  netbios-ssn Samba smbd 4
2222/tcp open  tcpwrapped
3333/tcp open  tcpwrapped
4444/tcp open  tcpwrapped
4701/tcp open  tcpwrapped
4703/tcp open  tcpwrapped
4704/tcp open  tcpwrapped
4705/tcp open  tcpwrapped
4802/tcp open  tcpwrapped
4803/tcp open  tcpwrapped
4804/tcp open  tcpwrapped
4805/tcp open  tcpwrapped
4901/tcp open  tcpwrapped
4902/tcp open  tcpwrapped
4903/tcp open  tcpwrapped
4904/tcp open  tcpwrapped
4905/tcp open  tcpwrapped
4950/tcp open  tcpwrapped
4951/tcp open  tcpwrapped
4952/tcp open  tcpwrapped
4953/tcp open  tcpwrapped
4954/tcp open  tcpwrapped
5555/tcp open  tcpwrapped
6666/tcp open  tcpwrapped
MAC Address: 00:0C:29:78:44:DC (VMware)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.72 seconds

┌──(kali㉿kali)-[~/ANSHIO]
└─$
```

OS Detection

```
kali@kali: ~/ANSHIO
File  Actions  Edit  View  Help
└─$ sudo nmap -O -p- 192.168.233.128
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-21 05:04 EDT
Nmap scan report for 192.168.233.128
Host is up (0.0015s latency).
Not shown: 65508 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
2222/tcp open  EtherNetIP-1
3333/tcp open  dec-notes
4444/tcp open  krb524
4701/tcp open  netxms-mgmt
4702/tcp open  netxms-sync
4703/tcp open  npqes-test
4704/tcp open  assuria-ins
4705/tcp open  unknown
4801/tcp open  iwec
4802/tcp open  ilss
4803/tcp open  notateit
4804/tcp open  aja-ntv4-disc
4805/tcp open  unknown
4901/tcp open  flr_agent
4903/tcp open  unknown
4904/tcp open  unknown
4905/tcp open  unknown
4950/tcp open  sybasesrvmon
4951/tcp open  pwgwims
4952/tcp open  sagxtsds
4953/tcp open  dbsyncarbiter
4954/tcp open  unknown
5555/tcp open  freeciv
6666/tcp open  irc
MAC Address: 00:0C:29:78:44:DC (VMware)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4)
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.10 seconds
```

Script Automation Scan



```
┌──(kali㉿kali)-[~/ANSHIO]
└─$ sudo nmap -A -p- 192.168.233.128
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-21 05:12 EDT
Nmap scan report for 192.168.233.128
Host is up (0.0010s latency).
Not shown: 65507 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
22/tcp   open  ssh         OpenSSH 9.6p1 Ubuntu 3ubuntu13.9 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 ab:82:af:ff:d3:f4:a0:2d:6f:35:02:b1:30:70:64:88 (ECDSA)
|_  256 7a:e5:c6:c2:6b:ec:95:48:c0:f0:30:53:ff:8a:ce:c9 (ED25519)
139/tcp  open  netbios-ssn Samba smbd 4
445/tcp  open  netbios-ssn Samba smbd 4
2222/tcp open  tcpwrapped
|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)
3333/tcp open  tcpwrapped
4444/tcp open  tcpwrapped
4701/tcp open  tcpwrapped
4702/tcp open  tcpwrapped
4703/tcp open  tcpwrapped
4704/tcp open  tcpwrapped
4705/tcp open  tcpwrapped
4801/tcp open  tcpwrapped
4802/tcp open  tcpwrapped
4803/tcp open  tcpwrapped
4804/tcp open  tcpwrapped
4805/tcp open  tcpwrapped
4901/tcp open  tcpwrapped
4902/tcp open  tcpwrapped
4903/tcp open  tcpwrapped
4904/tcp open  tcpwrapped
4905/tcp open  tcpwrapped
4950/tcp open  tcpwrapped
4951/tcp open  tcpwrapped
4952/tcp open  tcpwrapped
4953/tcp open  tcpwrapped
4954/tcp open  tcpwrapped
5555/tcp open  tcpwrapped
6666/tcp open  tcpwrapped
MAC Address: 00:0C:29:78:44:DC (VMware)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4)
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
| smb2-time:
|   date: 2025-04-21T09:13:07
|_  start_date: N/A
|_nbstat: NetBIOS name: UBUNTU, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
| smb2-security-mode:
|   3:1:1:
|_    Message signing enabled but not required

TRACEROUTE
HOP RTT     ADDRESS
1   1.04 ms 192.168.233.128

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 28.13 seconds

┌──(kali㉿kali)-[~/ANSHIO]
└─$
```

Vulnerability Script Scanning

```
kali@kali: ~/ANSHIO
File  Actions  Edit  View  Help

┌──(kali㉿kali)-[~/ANSHIO]
└─$ sudo nmap -p-  --script vuln 192.168.233.128
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-21 05:18 EDT
Nmap scan report for 192.168.233.128
Host is up (0.0028s latency).
Not shown: 65508 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
2222/tcp open  EtherNetIP-1
3333/tcp open  dec-notes
4444/tcp open  krb524
4701/tcp open  netxms-mgmt
4702/tcp open  netxms-sync
4703/tcp open  npqes-test
4704/tcp open  assuria-ins
4705/tcp open  unknown
4801/tcp open  iwec
4803/tcp open  notateit
4804/tcp open  aja-ntv4-disc
4805/tcp open  unknown
4901/tcp open  flr_agent
4902/tcp open  magiccontrol
4903/tcp open  unknown
4904/tcp open  unknown
4905/tcp open  unknown
4950/tcp open  sybasesrvmon
4951/tcp open  pwgwims
4952/tcp open  sagxtsds
4953/tcp open  dbsyncarbiter
4954/tcp open  unknown
5555/tcp open  freeciv
6666/tcp open  irc
| irc-botnet-channels:
|_  ERROR: EOF
MAC Address: 00:0C:29:78:44:DC (VMware)

Host script results:
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: ERROR: Server returned less data than it was supposed to (one or more f
ields are missing); aborting [9]
|_samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: ERROR: Server returned less data than it was supposed to (one or
 more fields are missing); aborting [9]
|_smb-vuln-ms10-054: false

Nmap done: 1 IP address (1 host up) scanned in 46.32 seconds
```

DNS Specific Scans

```
┌──(kali㉿kali)-[~/ANSHIO]
└─$ sudo nmap --script dns-zone-transfer -p 53 192.168.233.128
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-21 18:26 EDT
Nmap scan report for 192.168.233.128
Host is up (0.00066s latency).

PORT   STATE  SERVICE
53/tcp closed domain
MAC Address: 00:0C:29:78:44:DC (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.24 seconds

┌──(kali㉿kali)-[~/ANSHIO]
└─$ 
```

DHCP Discovery

```
┌──(kali㉿kali)-[~/ANSHIO]
└─$ sudo nmap --script broadcast-dhcp-discover 192.168.233.128
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-21 05:39 EDT
Pre-scan script results:
| broadcast-dhcp-discover:
|   Response 1 of 1:
|     Interface: eth0
|     IP Offered: 192.168.233.130
|     DHCP Message Type: DHCPOFFER
|     Server Identifier: 192.168.233.254
|     IP Address Lease Time: 30m00s
|     Subnet Mask: 255.255.255.0
|     Router: 192.168.233.2
|     Domain Name Server: 192.168.233.2
|     Domain Name: localdomain
|     Broadcast Address: 192.168.233.255
|     NetBIOS Name Server: 192.168.233.2
|     Renewal Time Value: 15m00s
|_    Rebinding Time Value: 26m15s
Nmap scan report for 192.168.233.128
Host is up (0.0021s latency).
Not shown: 992 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
2222/tcp open  EtherNetIP-1
3333/tcp open  dec-notes
4444/tcp open  krb524
5555/tcp open  freeciv
6666/tcp open  irc
MAC Address: 00:0C:29:78:44:DC (VMware)

Nmap done: 1 IP address (1 host up) scanned in 10.41 seconds
```

sudo nmap --script "dns-* and not dns-brute"

```
┌──(kali㉿kali)-[~/ANSHIO]
└─$ sudo nmap --script "dns-* and not dns-brute" 192.168.233.128
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-21 05:35 EDT
Nmap scan report for 192.168.233.128
Host is up (0.0012s latency).
Not shown: 992 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
2222/tcp open  EtherNetIP-1
3333/tcp open  dec-notes
4444/tcp open  krb524
5555/tcp open  freeciv
6666/tcp open  irc
MAC Address: 00:0C:29:78:44:DC (VMware)

Host script results:
| dns-blacklist:
|   SPAM
|_    l2.apews.org - FAIL

Nmap done: 1 IP address (1 host up) scanned in 5.47 seconds

┌──(kali㉿kali)-[~/ANSHIO]
└─$ sudo nmap --script dns-zone-transfer -p- 192.168.233.128
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-21 05:36 EDT
Nmap scan report for 192.168.233.128
Host is up (0.0023s latency).
Not shown: 65507 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
```

SMP VULNERABILITY SCAN

```
┌──(kali㉿kali)-[~/ANSHIO]
└─$ sudo nmap --script smb-enum-shares,smb-enum-users 192.168.233.128
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-21 05:41 EDT
Nmap scan report for 192.168.233.128
Host is up (0.0015s latency).
Not shown: 992 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
2222/tcp open  EtherNetIP-1
3333/tcp open  dec-notes
4444/tcp open  krb524
5555/tcp open  freeciv
6666/tcp open  irc
MAC Address: 00:0C:29:78:44:DC (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.39 seconds

┌──(kali㉿kali)-[~/ANSHIO]
└─$ 
```

Appendix C: System Configuration Analysis

DNS Configuration Issues - Logged into ssh port using ssh arturo@192.168.233.128

```
arturo@vm:~$ systemctl status bind9
Unit bind9.service could not be found.
```

```
arturo@vm:~$ ls -la /etc/bind/
total 56
drwxr-xr-x   2 root root 4096 Apr 16 00:36 .
drwxr-xr-x 104 root root 4096 Apr 16 00:23 ..
-rw-r--r--   1 root root 2403 Jan 28 14:26 bind.keys.dpkg-new
-rw-r--r--   1 root root  255 Sep 23  2024 db.0.dpkg-new
-rw-r--r--   1 root root  271 Sep 23  2024 db.127.dpkg-new
-rw-r--r--   1 root root  237 Sep 23  2024 db.255.dpkg-new
-rw-r--r--   1 root root  353 Sep 23  2024 db.empty.dpkg-new
-rw-r--r--   1 root root  270 Sep 23  2024 db.local.dpkg-new
-rw-r--r--   1 root root  498 Sep 23  2024 named.conf.default-zones.dpkg-new
-rw-r--r--   1 root root  458 Sep 23  2024 named.conf.dpkg-new
-rw-r--r--   1 root root   98 Apr 16 00:36 named.conf.local
-rw-r--r--   1 root root  165 Sep 23  2024 named.conf.local.dpkg-new
-rw-r--r--   1 root root  846 Sep 23  2024 named.conf.options.dpkg-new
-rw-r--r--   1 root root 1317 Sep 23  2024 zones.rfc1918.dpkg-new
```

```
arturo@vm:~$ cat /etc/bind/named.conf.local
zone "example.com" IN { type master; file "/etc/bind/db.example.com"; allow-transfer { any;
 }; };
arturo@vm:~$ 
```

DHCP Configuration Issues

```
arturo@vm:~$ systemctl status isc-dhcp-serve
Unit isc-dhcp-serve.service could not be found.
```

```
arturo@vm:~$ ls -la /etc/dhcp/
total 24
drwxr-xr-x   3 root root 4096 Apr 16 00:36 .
drwxr-xr-x 104 root root 4096 Apr 16 00:23 ..
drwxr-xr-x   2 root root 4096 Apr 14 17:37 dhclient-exit-hooks.d
-rw-r--r--   1 root root  158 Apr 16 00:36 dhcpd.conf
-rw-r--r--   1 root root 3646 Apr 22  2024 dhcpd.conf.dpkg-new
-rw-r--r--   1 root root 3331 Apr 22  2024 dhcpd6.conf.dpkg-new
arturo@vm:~$ cat /etc/dhcp/dhcpd.conf
option domain-name "example.com";
option domain-name-servers ns1.example.com;
subnet 192.168.0.0 netmask 255.255.255.0 { range 192.168.0.100 192.168.0.200; }
arturo@vm:~$
```

Running Services Analysis

```
arturo@vm:~$ systemctl list-units --type=service --state=running
  UNIT                          LOAD   ACTIVE SUB
DESCRIPTION
  dbus.service                  loaded active running D-Bus System Message Bus
  getty@tty1.service            loaded active running Getty on tty1
  multipathd.service            loaded active running Device-Mapper Multipath Device Controller
  nmbd.service                  loaded active running Samba NMB Daemon
  open-vm-tools.service         loaded active running Service for virtual machines hosted on VMware
  postgresql@16-main.service    loaded active running PostgreSQL Cluster 16-main
  redis-server.service          loaded active running Advanced key-value store
  smbd.service                  loaded active running Samba SMB Daemon
  ssh.service                   loaded active running OpenBSD Secure Shell server
  systemd-journald.service      loaded active running Journal Service
  systemd-logind.service        loaded active running User Login Management
  systemd-networkd.service      loaded active running Network Configuration
  systemd-resolved.service      loaded active running Network Name Resolution
  systemd-timesyncd.service     loaded active running Network Time Synchronization
  systemd-udevd.service         loaded active running Rule-based Manager for Device Events and Files
  unattended-upgrades.service   loaded active running Unattended Upgrades Shutdown
  user@1000.service             loaded active running User Manager for UID 1000
  vgauth.service                loaded active running Authentication service for virtual machines hosted on VMware
  xinetd.service                loaded active running Xinetd A Powerful Replacement For Inetd

Legend: LOAD   → Reflects whether the unit definition was properly loaded.
        ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
        SUB    → The low-level unit activation state, values depend on unit type.

19 loaded units listed.
```

Package Analysis

```
arturo@vm:~$ dpkg -l | grep -E 'bind|dhcp|dns'
iU  bind9                        1:9.18.30-0ubuntu0.24.04.2              amd64        Internet Domain Name Server
iU  bind9-libs:amd64             1:9.18.30-0ubuntu0.24.04.2              amd64        Shared Libraries used by BIND 9
iU  bind9-utils                  1:9.18.30-0ubuntu0.24.04.2              amd64        Utilities for BIND 9
ii  dhcpcd-base                  1:10.0.6-1ubuntu3.1                     amd64        DHCPv4 and DHCPv6 dual-stack client
(binaries and exit hooks)
iU  dns-root-data                2024071801~ubuntu0.24.04.1              all          DNS root hints and DNSSEC trust anch
or
iU  isc-dhcp-server              4.4.3-P1-4ubuntu2                       amd64        ISC DHCP server for automatic IP add
ress assignment
ii  libwbclient0:amd64           2:4.19.5+dfsg-4ubuntu9                  amd64        Samba winbind client library
rc  php8.3-mcrypt                3:1.0.5-4ubuntu1                        amd64        PHP bindings for the libmcrypt libra
ry
ii  python3-dnspython            2.6.1-1ubuntu1                          all          DNS toolkit for Python 3
ii  python3-gi                   3.48.2-1                                amd64        Python 3 bindings for gobject-intros
pection libraries
ii  python3-ldb                  2:2.8.0+samba4.19.5+dfsg-4ubuntu9       amd64        Python 3 bindings for LDB
ii  python3-netplan              1.1.1-1~ubuntu24.04.1                   amd64        Declarative network configuration Py
thon bindings
ii  python3-samba                2:4.19.5+dfsg-4ubuntu9                  amd64        Python 3 bindings for Samba
ii  python3-systemd              235-1build4                             amd64        Python 3 bindings for systemd
ii  python3-talloc:amd64         2.4.2-1build2                           amd64        hierarchical pool based memory alloc
ator - Python3 bindings
ii  python3-tdb                  1.4.10-1build1                          amd64        Python3 bindings for TDB
iU  rpcbind                      1.2.6-7ubuntu2                          amd64        converts RPC program numbers into un
iversal addresses
```

7. REFERENCES

- Nmap – 2025 - Nmap Reference Guide. https://nmap.org/book/man.html
- ISC DHCP – 2025 - DHCP Server Documents. https://www.isc.org/dhcp/
- OWASP – 2024 - Server Security Testing Guide. https://owasp.org/www-project-web-security-testing-guide/
- NIST (2024). https://csrc.nist.gov/publications/detail/sp/800-123/final